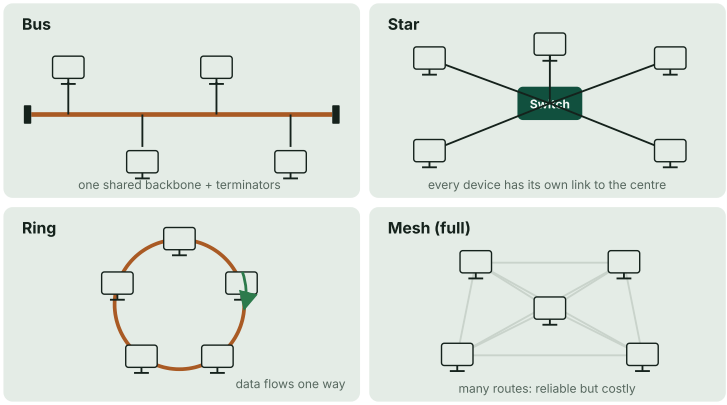
# Network types, topologies and hardware

IS09 FACT FILE

**Specification: students should be able to**

- Describe LAN, WAN, PAN and MAN networks
- Compare bus, star, ring and mesh topologies
- Describe the function of network hardware: NIC, switch, router, wireless access point and gateway

**Key words:** LAN, WAN, PAN, MAN, topology, bus, star, ring, mesh, NIC, switch, router, wireless access point, gateway




The four network topologies: bus with a shared backbone, star with a central switch, ring as a closed loop, and mesh with multiple links

## Network types

Networks are classified by the area they cover. A personal area network links a person's own devices over a few metres, often by Bluetooth. A local area network covers one site and is owned and managed by the organisation. A metropolitan area network covers a town or city, linking sites such as council buildings or university campuses. A wide area network spans countries, using telecommunications links leased from third parties; the internet is the largest WAN.

## Topologies

A topology is the arrangement of devices and links. A bus shares one backbone cable, so it is cheap but suffers collisions and fails if the backbone breaks. A ring passes data around a loop in one direction with no collisions, but one break can stop it. A star connects each device to a central switch, isolating cable faults and giving high performance, though the switch is a single point of failure. A mesh gives multiple routes, providing resilience at high cost.

## Hardware

A network interface card provides the physical connection and a unique MAC address. A switch forwards frames within a LAN to the correct port using MAC addresses. A router joins different networks and forwards packets between them using IP addresses and routing tables. A wireless access point connects wireless devices to a wired network. A gateway joins networks that use different protocols, translating between them.

## T Key terms from the mark schemes

**Local area network (LAN)**
A network covering a single site or building, with infrastructure owned and managed by the organisation.

**Wide area network (WAN)**
A network covering a large geographical area, connecting LANs using third-party telecommunications links.

**Topology**
The physical or logical arrangement of the devices and links in a network.

**Switch**
A device that connects devices within a LAN and forwards each frame only to the port of its destination, using MAC addresses.

**Router**
A device that connects different networks and forwards packets towards their destination using IP addresses and routing tables.

**Gateway**
A device that connects two networks using different protocols and translates between them.

### Exam focus

Many questions ask you to recommend and justify a topology or device for a scenario. A justification needs a feature and its benefit in context, for example 'a star topology means a cable fault in one guest room only affects that room, so the rest of the hotel stays online'. When comparing, use the same criteria for each option (reliability, performance, cost, scalability). Do not describe a switch as broadcasting data – that is a hub – and remember routers use IP addresses while switches use MAC addresses.

## ? Check yourself

1. State one difference between a LAN and a WAN. [1]
2. Describe the function of a switch in a star network. [2]
3. Explain the difference between a router and a gateway. [2]

**Answers:** 1. Any one from: a LAN covers a small area such as one site or building, while a WAN covers a large geographical area such as countries (1) / a LAN's infrastructure is owned by the organisation, while a WAN uses third-party telecoms links / leased lines (1). 2. The switch connects all the devices in the star; each device has its own cable to it (1) / it learns the MAC address of each device and forwards each frame only to the port of the destination device, rather than to all devices, reducing traffic and collisions (1). 3. A router connects different networks (e.g. a LAN to the internet) and forwards packets using IP addresses and routing tables to choose the best route (1) / a gateway connects networks that use different protocols, translating between them (1).